Assignment 2 : MA20268

Disclaimer: This solution to assignments is made (typed & thought upon) completely by hand and with full understanding of the content by me

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Constraints: Using only `numpy` *for calculations and* `matplotlib` *for visualizations*

Question 1:

1. Generate two vectors $u \in \mathbb{R}^{200}$ and $v \in \mathbb{R}^{200}$ sampled independently from a uniform random distribution on the range $[-5, 5]$.

   Define the target variable $y \in \mathbb{R}^{200}$ by

   $$y_i = \begin{cases} 1, & \text{if } u_i^2 + v_i^2 > 4, \\ 0, & \text{otherwise,} \end{cases} \qquad i = 1, \ldots, 200. \tag{1}$$

   Here, y_i, u_i, and v_i denote the i-th entries of the vectors y, u, and v, respectively.

   Now, define the following basis functions:

   $$\begin{aligned} g_0(u_i, v_i) &= 1, \qquad g_1(u_i, v_i) = u_i, \qquad g_2(u_i, v_i) = v_i, \\ g_3(u_i, v_i) &= u_i^2, \qquad g_4(u_i, v_i) = v_i^2, \qquad g_5(u_i, v_i) = u_i v_i. \end{aligned} \tag{2}$$

   Use the basis functions to define the matrix $C \in \mathbb{R}^{200 \times 6}$ where

   $$C_i^j = g_j(u_i, v_i), \tag{3}$$

   with i as the row index and j as the column index.

   Use the least-squares method to predict the target variable y by learning the coefficient vector $w \in \mathbb{R}^6$ in the linear model

   $$y \approx Cw. \tag{4}$$

Figure 1: Q1

Initializing $u, v \in \mathbb{R}^{200}$

```
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

u = np.random.uniform(-5, 5, 200)
v = np.random.uniform(-5, 5, 200)
```

Now, the target variable $y \in \mathbb{R}^{200}$ as defined:

```
y = np.where(u**2 + v**2 > 4, 1, 0)
```

Given basis functions: $g_0(u_i, v_i) = 1$, $g_1(u_i, v_i) = u_i$, $g_2(u_i, v_i) = v_i$, $g_3(u_i, v_i) = {u_i}^2$, $g_4(u_i, v_i) = {v_i}^2$, $g_5(u_i, v_i) = v_i u_i$

```
def g1(ui, vi):
    return 1

def g2(ui, vi):
    return ui

def g3(ui, vi):
    return vi

def g4(ui, vi):
    return ui**2

def g5(ui, vi):
    return vi**2

def g6(ui, vi):
    return ui*vi

basis_func = {
    0: g1,
    1: g2,
    2: g3,
    3: g4,
    4: g5,
    5: g6
}
```

Defining matrix $C \in \mathbb{R}^{200\times 6}$

```
c = np.zeros((200, 6))

for i in range(200):
    for j in range(6):
        c[i][j] = basis_func.get(j, lambda u,v: None)(u[i], v[i])
```

Training for $\tilde{y}$:

the model for this problem is:

$\tilde{y}(u, v) = \sum_{i=0}^{5} w_i g_i(u, v) = w_0 + w_1 u + w_2 v + w_3 u^2 + w_4 v^2 + w_5 uv$

That is, $\mathbf{y} = C\mathbf{w}$

to use least squares method, define: $E(\mathbf{w}) = \frac{1}{2} \sum_{i=1}^{200} (y_i - \tilde{y}_i)^2$

Now, our main goal is to minimize $E(\mathbf{w})$.

to get $\tilde{w}$ we solve the least squares problem $\min_w \|\mathbf{y} - C\mathbf{w}\|^2$

```
w = np.linalg.lstsq(c, y, rcond=None)[0]

print("Weights: ")
for i, val in enumerate(w):
    print(i+1, val)
```

```
Weights:
1 0.7033078960988596
2 0.006958867604766851
3 0.0022148505318308467
4 0.013205693000868256
5 0.012235367790864862
6 -0.0001298706940927332
```

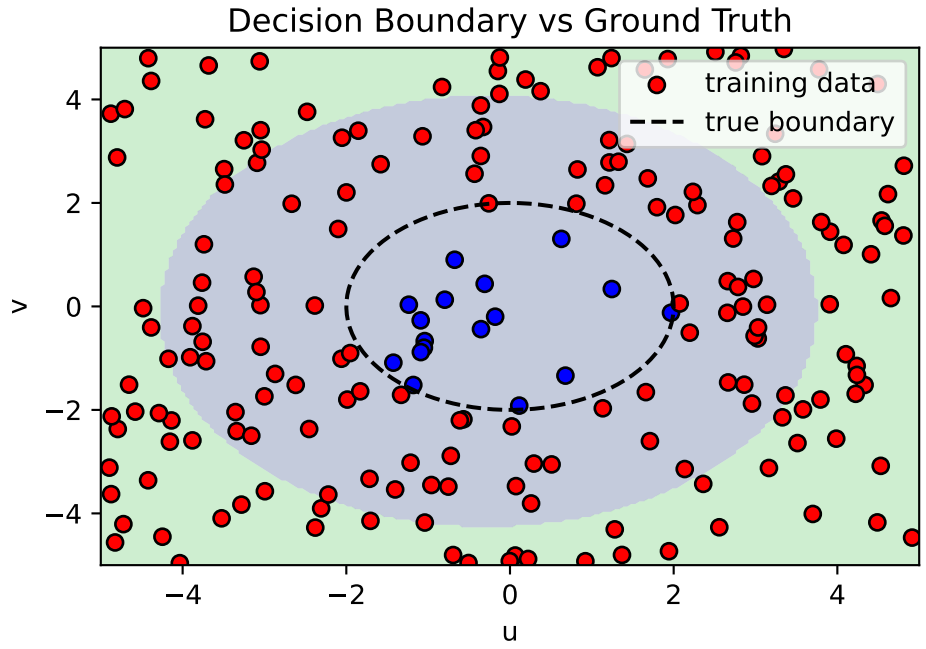
Final views on the problem

The target set and basis functions used here trained the model $\tilde{y}$ to predict whether a point (u, v) lies outside the circle of radius '2' or not (defined by the target condition: $u^2 + v^2 > 4$). So, now we should be able to predict this using the approximated weights $\tilde{w}$.

Visualization: True Boundary vs. Predicted model

```
# helper function for predictions
def pred(u, v, w):
    features = np.array([1,u,v,u**2,v**2,u*v])
    y_hat = features @ w # dot product of given surface with w
    return y_hat

threshold = np.mean(y)


# create a grid for visualization
grid_size = 200
u_grid = np.linspace(-5, 5, grid_size)
v_grid = np.linspace(-5, 5, grid_size)
U, V = np.meshgrid(u_grid, v_grid)


# compute predictions over grid
Z = np.zeros_like(U)

for i in range(grid_size):
    for j in range(grid_size):
        Z[i, j] = pred(U[i, j], V[i, j], w)

# decision boundary using threshold (chosen arbitrarily as the mean)
Z_class = (Z > threshold).astype(int)

plt.figure()

# plot decision regions
plt.contourf(U, V, Z_class, alpha=0.3, levels=[-0.1, 0.5, 1.1])
plt.scatter(u, v, c=y, cmap='bwr', edgecolors='k', label="training data")
```

```
# plot true boundary (circle of radius 2)
theta = np.linspace(0, 2*np.pi, 200)
plt.plot(2*np.cos(theta), 2*np.sin(theta), 'k--', label="true boundary")

plt.xlabel("u")
plt.ylabel("v")
plt.legend()
plt.title("Decision Boundary vs Ground Truth")

plt.show()
```



Inference to this image : ....

Testing

```
TEST_SIZE = 100
MAX_PRINT = 10
u_test = np.random.uniform(-5, 5, TEST_SIZE)
v_test = np.random.uniform(-5, 5, TEST_SIZE)
print("TESTING on random uniform test with size: ", TEST_SIZE )

correct = 0

for i in range(TEST_SIZE):
    y_hat = pred(u_test[i], v_test[i], w)
    prediction = 1 if y_hat > threshold else 0
    t = 1 if u_test[i]**2 + v_test[i]**2 > 4 else 0

    if prediction == t:
        correct += 1
    if i < MAX_PRINT:
        print(f"({u_test[i]:.2f}, {v_test[i]:.2f}) => model: {prediction} truth: {t}")
    elif i == MAX_PRINT:
        print("...")
```

```
print(f"points outside the circle in training data: {threshold}%") # shows skewness in the
↪  random training dataset
print(f"accuracy: {(correct/TEST_SIZE)*100}%")
```

```
TESTING on random uniform test with size:  100
(-2.33, 2.65) => model: 0 truth: 1
(-3.13, 4.89) => model: 1 truth: 1
(-4.02, -0.07) => model: 0 truth: 1
(0.95, 1.48) => model: 0 truth: 0
(-1.26, -0.10) => model: 0 truth: 0
(3.59, -3.02) => model: 1 truth: 1
(-4.34, 0.05) => model: 1 truth: 1
(3.45, 4.35) => model: 1 truth: 1
(-2.17, 3.10) => model: 0 truth: 1
(-3.95, 4.70) => model: 1 truth: 1
...
points outside the circle in training data: 0.915%
accuracy: 63.0%
```

Note: Since the dataset is not balanced (i.e., the proportion of class 1 labels is significantly higher than that of class 0), a threshold of 0.5 is not appropriate. As the model is obtained via least-squares regression and produces continuous outputs, a decision threshold is required for classification. In this case, we use the empirical mean of the training labels as the threshold to account for the class imbalance.

Question 2:

2. Download the Fashion-MNIST dataset (both train and test sets, 28×28 images). Crop each image by 4 pixels from both rows and columns (resulting in $20 \times 20 = 400$ dimensions per vectorized image).

   Let x_i be a sample in the train set and $c(x_i)$ be its true class label (0–9). Run 9 separate simulations comparing class 0 (T-shirt/top) against each of the other classes, where:

   - $c(x_i) = 0 \Rightarrow$ label $+1$

   - $c(x_i) = k \ (k \neq 0) \Rightarrow$ label -1

   The 9 experiments are:

$$\begin{array}{lll} 0 \text{ vs } 1, & 0 \text{ vs } 2, & 0 \text{ vs } 3, \\ 0 \text{ vs } 4, & 0 \text{ vs } 5, & 0 \text{ vs } 6, \\ 0 \text{ vs } 7, & 0 \text{ vs } 8, & 0 \text{ vs } 9 \end{array} \tag{5}$$

   For each experiment, use least squares to train a linear regression model $f(\cdot)$ on the binary train subset.

   On the test set, predict:

$$\hat{c}(x_j) = \begin{cases} +1 & \text{if } f(x_j) > 0 \\ -1 & \text{otherwise} \end{cases} \tag{6}$$

   Using ground truth test labels, compute the misclassification error for samples truly belonging to class 0 or the opposing class k.

Setting up MNIST dataset

using keras datasets for clean mnist import to avoid messy mnist_reader util files (alternative way was to use zalando's mnist_reader util)

```
from tensorflow.keras.datasets import fashion_mnist
(x_train, y_train), (x_test, y_test) = fashion_mnist.load_data()
```

To reduce dimensionality and remove irrelevant border pixels, we crop image from 28×28 to 20×20:

```
x_train = x_train[:, 4:24, 4:24]
x_test = x_test[:, 4:24, 4:24]
```

Each image is flattened into a feature vector:

```
x_train = x_train.reshape(len(x_train), -1)
x_test = x_test.reshape(len(x_test), -1)

# normalize pixel values to [0,1] to improve numerical stability of least squares
x_train = x_train / 255.0
x_test = x_test / 255.0
```

Setting up the Model

First, we define a helper function to *extract only the relevant classes (0 and k) and relabel them*:

```
# Helper function to filter out x, y sets into two label classes 0 and k
def get_binary(x ,y, k):
    mask = (y==0) | (y==k)
    x_bin = x[mask]
    y_bin = y[mask]

    y_bin = np.where(y_bin == 0, 1, -1);

    return x_bin, y_bin
```

Model: Least Square Classifier

We model the prediction as: $\tilde{y} = \mathbf{w}^T\mathbf{x} + b$, where b is the bias term.

To incorporate the bias term, we augment the feature matrix with a column of ones. The opimal weight $\tilde{w}$ are obtained by solving the equation:

$\min_{\mathbf{w}} |X\mathbf{w} - y|^2$

Using the Moore-Penrose pseudoinverse (built-in in numpy):

```
def train_ls(X, y):
    X = np.c_[np.ones(X.shape[0]), X]  # add bias
    w = np.linalg.pinv(X) @ y # best fit line using pseudoinverse
    return w
```

Prediciton Rule:

Predictions are made using the sign of the model output:

```
def predict(X, w):
    X = np.c_[np.ones(X.shape[0]), X]
    return np.where(X @ w > 0, 1, -1)
```

Evaluation Metric:

We use the misclassification error defined as: Error $= \frac{1}{N}\sum_{i=1}^{N} \mathbb{1}(y_i \neq \hat{y}_i)$

```
def misclassification_error(y_true, y_pred):
    return np.mean(y_true != y_pred)
```

Running Experiments

Training and evaluating the model for each $k \in 1, \ldots, 9$:

```
results = {}

for k in range(1, 10):
```

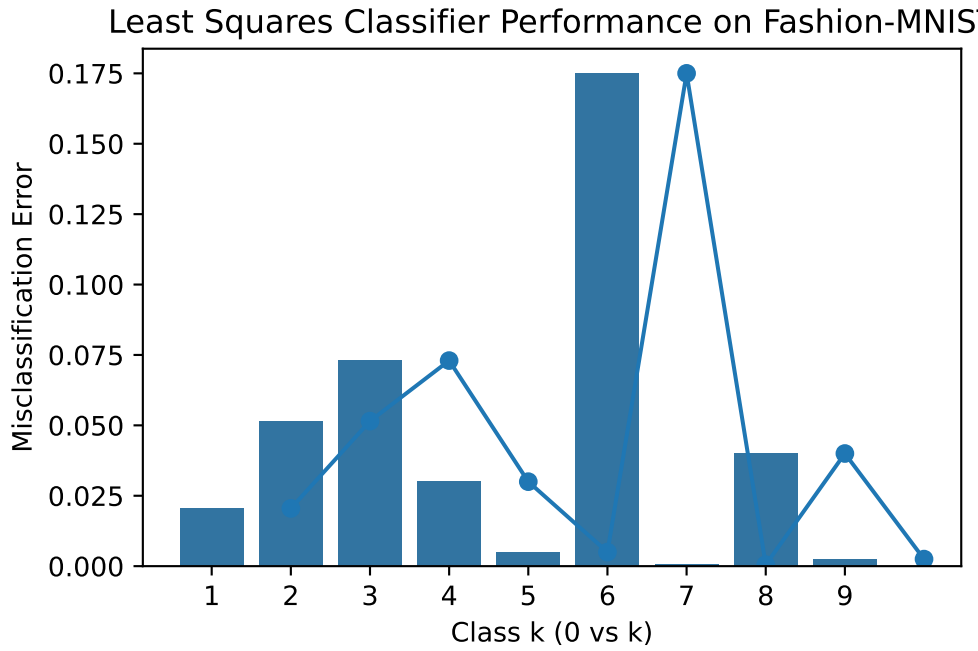
```
    X_train_k, y_train_k = get_binary(x_train, y_train, k)
    w = train_ls(X_train_k, y_train_k)
    X_test_k, y_test_k = get_binary(x_test, y_test, k)
    y_pred = predict(X_test_k, w)
    error = misclassification_error(y_test_k, y_pred)

    results[k] = error
    print(f"0 vs {k} => error: {error:.4f}")
```

```
0 vs 1 => error: 0.0205
0 vs 2 => error: 0.0515
0 vs 3 => error: 0.0730
0 vs 4 => error: 0.0300
0 vs 5 => error: 0.0050
0 vs 6 => error: 0.1750
0 vs 7 => error: 0.0005
0 vs 8 => error: 0.0400
0 vs 9 => error: 0.0025
```

Interepretation and Inference

Visualization

```
# convert results to lists
ks = list(results.keys())
errors = list(results.values())

plt.figure()

# bar plot
sns.barplot(x=ks, y=errors)

# optional: overlay line for trend
plt.plot(ks, errors, marker='o')

plt.xlabel("Class k (0 vs k)")
plt.ylabel("Misclassification Error")
plt.title("Least Squares Classifier Performance on Fashion-MNIST")

plt.show()
```



This experiment highlights how well a linear model trained via least squares can separate different classes in a high-dimensional image space.

Key observations:

- Since the model is linear, performance depends on how linearly separable the classes are.
- Some class pairs (e.g., visually distinct clothing types) are expected to yield lower error.
- Others with similar shapes or textures may result in higher misclassification.

Despite its simplicity, this approach provides a useful baseline before moving to more expressive models like logistic regression or neural networks.

Question 3:

3. Generate two vectors $u \in \mathbb{R}^{100}$ and $v \in \mathbb{R}^{100}$ sampled independently from a uniform random distribution on the range $[-5, 5]$.

   Define the target variable $y \in \mathbb{R}^{100}$ by

$$y_i = \begin{cases} +1 & \text{if } u_i v_i > 0.5, \\ -1 & \text{otherwise,} \end{cases} \tag{7}$$

   where y_i, u_i, v_i denote the i-th entries of y, u, v, respectively.

   Now define the following basis functions:

$$\begin{aligned} &g_0(u_i, v_i) = 1, \quad g_1(u_i, v_i) = u_i, \quad g_2(u_i, v_i) = v_i, \\ &g_3(u_i, v_i) = u_i^2, \quad g_4(u_i, v_i) = v_i^2, \quad g_5(u_i, v_i) = u_i v_i. \end{aligned} \tag{8}$$

   Use these to form the matrix $C \in \mathbb{R}^{100 \times 6}$ with entries $C_i^j = g_j(u_i, v_i)$, where i is the row index and j is the column index.

   Use least squares to learn a coefficient vector $w \in \mathbb{R}^6$ in the linear model $y \approx Cw$. Let $\hat{y} = Cw$ be the predictions, and define the Mean Squared Error (MSE) as

$$\text{MSE} = \sum_{i=1}^{100} (y_i - \hat{y}_i)^2. \tag{9}$$

   Then compute MSE.

Figure 2: Q3

Process

Our goal is to find MSE for the target variable $y \in \mathbb{R}^{100}$ given by:

$$y_i = \begin{cases} +1 & \text{if} u_i v_i > 0.5 \\ -1 & \text{otherwise} \end{cases}$$

i.e. a prediction model to show if a point (u,v) $\in$ {set of points outside hyprbola uv = 0.5 for $u \in \mathbf{u}, v \in \mathbf{v}$}

The dataset for training is *random points sampled independently from the uniform distribution* of points in a square of side length 5.

Defining the Training Dataset

```
Q3_SAMPLE_SIZE = 100
u = np.random.uniform(-5, 5, Q3_SAMPLE_SIZE)
v = np.random.uniform(-5, 5, Q3_SAMPLE_SIZE)
```

and the target variable **y** is:

```
y = np.where(u*v > 0.5, 1, -1)
```

Building the $\mathbb{C}$ matrix with given basis functions;

```
c = np.zeros((Q3_SAMPLE_SIZE, 6))

# using the basis function dictionary defined in Q1
for i in range(Q3_SAMPLE_SIZE):
    for j in range(6):
        c[i][j] = basis_func.get(j, lambda u,v: None)(u[i], v[i])
```

Solving for the optimal weights

The optimal weights $\mathbf{w}$ such that $\mathbb{C}\mathbf{w} \approx \mathbf{y}$ is calculated as:

```
w = np.linalg.lstsq(c, y, rcond=None)[0]

print("weights: ")
for i, val in enumerate(w):
    print(f"{i} => {val}")
```

```
weights:
0 => -0.24638026217861658
1 => -0.0015754191660804773
2 => 0.034441088111060454
3 => 0.008123862188397467
4 => 0.013099401880537832
5 => 0.09178428828788705
```

Calculating MSE

Here, we have $\hat{y} = \mathbb{C}\mathbf{w}$ and MSE is given by:

MSE $= \sum_{i=1}^{100}(y_i - \hat{y}_i)^2$

```
y_hat = c @ w
mse = np.sum((y-y_hat)**2)

print(f"mse = {mse:.5f}, mean: {np.mean((y-y_hat)**2)}")
```

```
mse = 46.14900, mean: 0.46148998414461184
```

Inference and Observations

- The model approximates a nonlinear boundary ($uv = 0.5$) using linear least squares with basis functions.
- Performance depends on whether the basis includes interaction terms (e.g., $\mathbf{uv}$).
- A lower MSE indicates better approximation of the nonlinear relationship.
- This is effectively a regression-based classification, where predictions are thresholded.

Question 4:

4. Generate a vector $r \in \mathbb{R}^{100}$ sampled from a uniform random distribution on $[-8, 8]$. Define the target variable $t \in \mathbb{R}^{100}$ by

$$t_i = e^{0.1 r_i} \cos(r_i), \tag{10}$$

where t_i, r_i denote the i-th entries of t, r.

Define the first set of basis functions:

$$\begin{aligned} h_0(r_i) = 1, \quad h_1(r_i) = r_i, \quad h_2(r_i) = r_i^2, \\ h_3(r_i) = r_i^3, \quad h_4(r_i) = r_i^4, \quad h_5(r_i) = r_i^5. \end{aligned} \tag{11}$$

Form the matrix $B \in \mathbb{R}^{100 \times 6}$ with $B_i^j = h_j(r_i)$. Use least squares to fit coefficients $\beta \in \mathbb{R}^6$ minimizing $\|B\beta - t\|_2^2$, and compute

$$\text{MSE}_1 = \frac{1}{100} \sum_{i=1}^{100} (t_i - \hat{t}_i)^2. \tag{12}$$

Now add three higher-order basis functions:

$$h_6(r_i) = r_i^6, \quad h_7(r_i) = r_i^7, \quad h_8(r_i) = r_i^8, \tag{13}$$

forming $B' \in \mathbb{R}^{100 \times 9}$ and coefficients $\beta' \in \mathbb{R}^9$. Compute MSE_2 similarly.

What happens to the MSE when adding the three new basis functions? (Hint: Plot predictions vs. ground truth to understand overfitting behavior.)

Figure 3: Q4

Setting up

Making the dataset:

```
r = np.random.uniform(-8,8,100)

t = np.exp(0.1*r) * np.cos(r)

B1 = np.vander(r, 6, increasing=True)

beta1 = np.linalg.lstsq(B1, t)[0]

t_hat1 = B1 @ beta1

mse1 = np.mean((t-t_hat1)**2)

print(f"mse: {mse1}")

B2 = np.vander(r, 9, increasing=True)
```

```
beta2 = np.linalg.lstsq(B2, t)[0]

t_hat2 = B2 @ beta2

mse2 = np.mean((t-t_hat2)**2)

print(f"mse2: {mse2}")
```

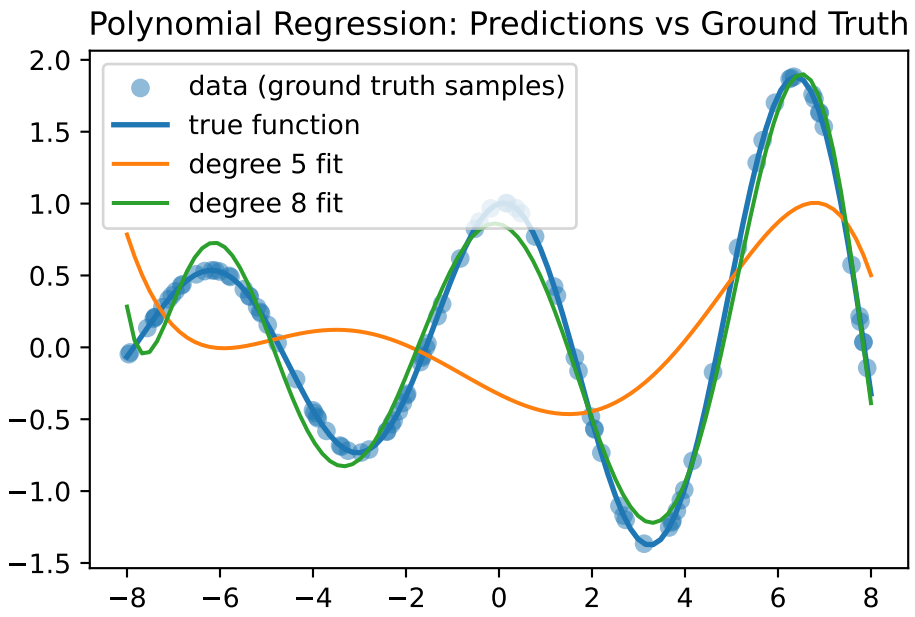
```
mse: 0.4680228593193963
mse2: 0.016282581226522493
```

Visualization: Truth vs. Predictions

For Ground truth: taking evenly spread numbers for getting the actual function

Predictions: using the order 6 and 8 models to plot separate graphs.

```
r_plot = np.linspace(-8, 8, 100)
t_true = np.exp(0.1 * r_plot) * np.cos(r_plot)

# plot predictions
B1_plot = np.vander(r_plot, 6, increasing=True)
t_pred1 = B1_plot @ beta1

B2_plot = np.vander(r_plot, 9, increasing=True)
t_pred2 = B2_plot @ beta2

# plot
plt.figure()

plt.scatter(r, t, label="data (ground truth samples)", alpha=0.5)
plt.plot(r_plot, t_true, label="true function", linewidth=2)
plt.plot(r_plot, t_pred1, label="degree 5 fit")
plt.plot(r_plot, t_pred2, label="degree 8 fit")

plt.legend()
plt.title("Polynomial Regression: Predictions vs Ground Truth")
plt.show()
```


Polynomial Regression: Predictions vs Ground Truth
data (ground truth samples)
true function
degree 5 fit
degree 8 fit
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−8
−6
−4
−2
0
2
4
6
8

Question 5:

5. Download the MNIST dataset (both train and test sets, 28×28 images). Crop each image by 5 pixels from both rows and columns (resulting in $18 \times 28 = 504$ dimensions), then standardize each sample by subtracting the mean and dividing by the variance across all pixels.

   Let x_i be a sample in the train set and $c(x_i)$ its true class label (0–9). Run 9 separate simulations comparing class 1 against each other class, where:

   - $c(x_i) = 1 \Rightarrow$ label $+1$
   - $c(x_i) = k$ $(k \neq 1) \Rightarrow$ label -1

   The experiments are:

$$\begin{aligned} &1 \text{ vs } 0, \quad 1 \text{ vs } 2, \quad 1 \text{ vs } 3, \\ &1 \text{ vs } 4, \quad 1 \text{ vs } 5, \quad 1 \text{ vs } 6, \\ &1 \text{ vs } 7, \quad 1 \text{ vs } 8, \quad 1 \text{ vs } 9 \end{aligned} \tag{14}$$

   For each experiment:

   (a) Train linear regression $f(\cdot)$ using least squares on the binary train subset

   (b) On test set, predict: $\hat{c}(x_j) = +1$ if $f(x_j) > 0$, else -1

   (c) Count misclassifications using ground truth test labels (only for true class 1 or k)

   When using standardization as preprocessing (vs raw pixels), what effect do you observe?

Figure 4: Q5

Setting up MNIST

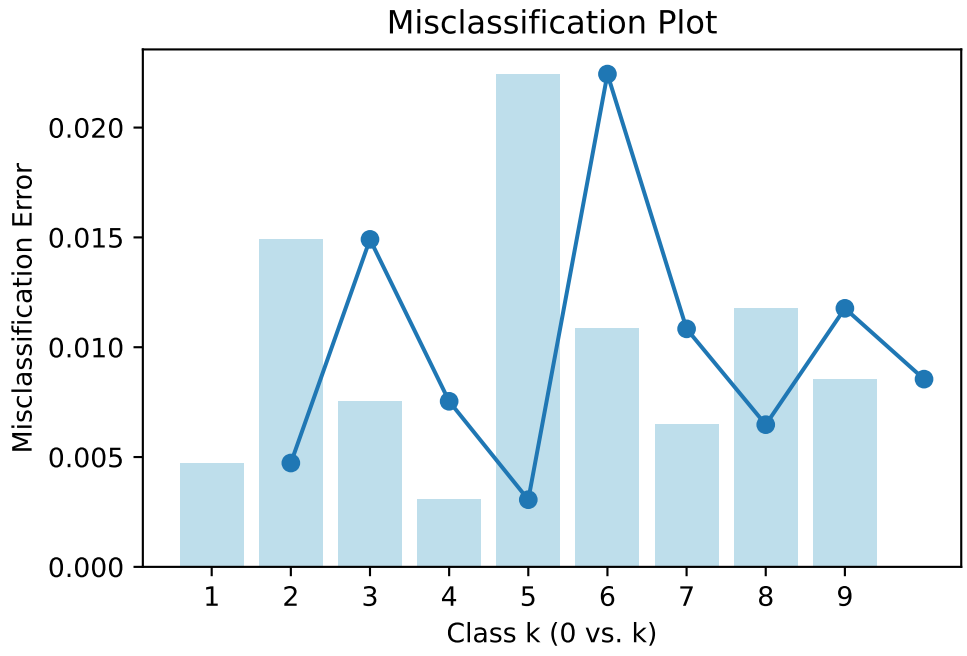
Setting up and finding the result:

```
from tensorflow.keras.datasets import mnist
(x_train, y_train), (x_test, y_test) = mnist.load_data()


# reducing dimensions
x_test = x_test[:, 5:23, 5:23]
x_train = x_train[:, 5:23, 5:23]

# flattening

x_train = x_train.reshape(len(x_train), -1)
```

```
x_test = x_test.reshape(len(x_test), -1)

# standardization
print(f"samples: {x_train.shape[0]}, features: {x_train.shape[1]}")
# print(x_train[0])
mean = x_train.mean(axis=0)
# print("mean: ", mean)
# sqsum = np.zeroes(x_train.shape[1])
# variance = np.zeros(x_train.shape[1])
# for i in range(x_train.shape[0]):
#    variance += (x_train[i] - mean)**2

# variance /= x_train.shape[0]

# print("varaince: ", variance.shape)

std_deviation = x_train.std(axis=0)

# print("std: ", std_deviation)

# finally,
x_train = (x_train - mean) / std_deviation
# print("standardized: ", x_train[0])
# print(x_train.mean(axis=0))   # should be ~0
# print(x_train.std(axis=0))    # should be ~1

# Helper function to filter out x, y sets into two label classes 0 and k
def get_binary(x ,y, k):
    mask = (y==0) | (y==k)
    x_bin = x[mask]
    y_bin = y[mask]

    y_bin = np.where(y_bin == 0, 1, -1);

    return x_bin, y_bin

def train_ls(x, y):
    x = np.c_[np.ones(x.shape[0]), x]
    return np.linalg.pinv(x) @ y

def predict(x, w):
    x = np.c_[np.ones(x.shape[0]), x]
    return np.where(x@w > 0, 1, -1)

def misclassification_error2(y_true, y_pred):
    return np.mean(y_true != y_pred)
for k in range(1, 10):
    x_train_k, y_train_k = get_binary(x_train, y_train, k)
    x_test_k, y_test_k = get_binary(x_test, y_test, k)
    w = train_ls(x_train_k, y_train_k)
    y_pred = predict(x_test_k, w)
    # print(y_pred)
    error = misclassification_error2(y_test_k, y_pred)
    results[k] = error

meanout = np.array(list(results.values()))
meanout = meanout.mean()
```

```
# print(results)
print(f"Mean misclassification error:  {meanout:.3f}")

# visualization
ks = list(results.keys())
errors = list(results.values())

plt.figure()

sns.barplot(x=ks, y=errors, color='skyblue', alpha=0.6)

## trend plot
plt.plot(ks, errors, marker='o')


plt.xlabel("Class k (0 vs. k)")
plt.ylabel("Misclassification Error")
plt.title("Misclassification Plot")

plt.show()


# print("weights: ", w)
```

```
samples: 60000, features: 324
Mean misclassification error:  0.010
```



New Things Learned:

@TODO: Q1 visual inference explanation and views update krlo @TODO: Q4 context and explanation likhna hai (code done)

Python code:

- `np.linalg.lstsq(A, b, rcond=None)` => solves the matrix linear equation directly: $A\mathbf{x} = b$
- `np.where(<condition>, <val_T>, <val_F>)` => allowed me to construct np vectors directly using conditions

Inference of problems:

Question 1:

- understood the differences in applications between regression and classification models
- got more familiar with terminologies like class labels
- understood the use of model and basis function in the QUERY => PREDICTION flow.

Question 2:

- Kuchh to seekha hu lekin likhna bhul gya .... ^_^'